

NUNN 7 NNE, CO, USA

WMO: 745340

Lat: 40.8066N

Lon: 104.7552W

Elev: 1643

StdP: 83.08

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94074

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.5	-16.9	-24.3	0.5	-19.7	-20.9	0.7	-13.8	12.0	0.0	10.9	0.3	1.7	N/A	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	34.0	14.9	32.5	14.8	30.9	14.8	17.9	26.4	17.3	26.1	16.6	25.8	3.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	13.6	19.6	14.8	12.9	19.2	13.9	12.1	18.9	57.7	26.4	55.2	25.9	53.1	26.0	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	7.7	-26.9	36.7	3.0	1.2	-29.1	37.6	-30.9	38.3	-32.6	39.0	-34.8	39.8		
(5)			-27.1	19.8	3.6	0.8	-29.7	20.4	-31.8	20.8	-33.8	21.2	-36.5	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-1.9	-1.7	3.9	7.1	11.9	18.2	21.6	20.5	16.1	8.7	2.6	-2.9
	DBStd	9.89	6.19	6.43	5.32	4.73	4.60	3.46	2.58	2.79	4.05	5.30	5.78	6.29
	HDD10.0	1757	370	329	197	107	35	1	0	0	5	86	227	400
	HDD18.3	3759	628	562	448	336	203	43	6	13	88	300	473	658
	CDD10.0	1289	1	1	7	21	93	246	361	325	187	44	4	0
	CDD18.3	250	0	0	0	0	2	39	108	80	21	0	0	0
	CDH23.3	4443	0	0	3	17	144	829	1594	1261	541	53	1	0
	CDH26.7	1862	0	0	0	1	25	342	758	542	189	5	0	0
Wind	WSAvg	3.0	3.0	3.3	3.5	3.9	3.3	2.8	2.5	2.5	2.6	2.9	2.8	2.9
Precipitation	PrecAvg	358	8	9	23	35	59	52	49	39	33	24	15	8
	PrecMax	479	24	18	92	88	151	116	93	133	142	61	33	34
	PrecMin	216	1	1	1	3	25	15	18	4	2	1	5	1
	PrecStd	69	6	5	22	22	32	28	21	24	28	16	8	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.6	23.4	25.6	29.4	35.5	35.8	34.8	33.1	27.4	21.8	15.4
		MCWB	5.2	6.4	8.5	9.9	13.3	14.4	15.2	14.4	13.8	11.6	7.8	4.9
	2%	DB	13.2	14.6	20.3	23.0	26.9	33.0	34.2	33.0	30.6	24.9	18.8	12.5
		MCWB	3.9	4.5	7.1	9.1	11.9	13.9	15.4	14.7	13.2	10.8	7.3	3.7
	5%	DB	10.5	11.7	17.6	20.4	24.9	30.5	32.9	31.6	29.0	22.4	16.0	9.9
		MCWB	2.9	3.6	6.2	8.3	11.5	14.4	15.6	14.5	12.7	9.5	6.0	2.3
	10%	DB	7.7	8.7	14.7	17.7	22.1	28.3	31.2	30.0	26.8	19.4	13.0	6.9
		MCWB	1.5	1.9	5.1	7.4	11.3	14.3	15.6	14.5	12.7	8.6	4.7	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	6.9	9.0	10.8	15.5	18.7	19.2	18.3	17.1	12.8	8.5	5.3
		MCDB	15.0	17.9	21.9	22.2	22.8	27.1	27.3	25.9	26.2	22.5	19.2	14.8
	2%	WB	4.2	5.1	7.8	9.8	13.9	17.2	18.3	17.2	15.8	11.7	7.4	4.0
		MCDB	12.2	14.5	19.4	20.1	20.9	25.8	26.8	26.3	24.9	21.3	17.9	12.0
	5%	WB	3.1	3.9	6.7	8.9	12.8	16.2	17.6	16.6	14.7	10.5	6.1	2.7
		MCDB	10.2	12.1	17.0	18.4	20.8	25.2	26.3	26.1	23.3	19.4	15.0	9.6
	10%	WB	1.9	2.4	5.5	7.9	11.7	15.3	17.0	15.8	13.9	9.3	4.8	1.3
		MCDB	7.9	9.1	14.3	16.7	19.4	24.3	25.9	25.7	22.8	18.0	12.5	6.9
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.6	15.9	15.2	14.7	16.9	17.2	17.3	17.0	15.8	15.3	13.9
		MCDBR	17.8	18.6	20.7	21.0	19.8	21.6	20.6	20.3	21.0	20.9	19.4	17.5
		MCWBR	10.8	10.4	9.9	9.2	7.9	7.5	5.8	6.3	7.5	9.0	9.9	10.3
	5% WB	MCDBR	17.8	18.4	19.9	19.1	16.5	18.1	16.3	17.5	16.5	18.0	18.8	17.2
		MCWBR	10.6	10.2	9.7	8.8	7.7	7.4	5.9	6.0	6.6	8.2	9.7	10.4
Clear-Sky Solar Irradiance	taub		0.239	0.246	0.269	0.295	0.320	0.324	0.352	0.345	0.308	0.269	0.252	0.242
	taud		2.594	2.585	2.528	2.479	2.466	2.477	2.437	2.447	2.532	2.620	2.577	2.569
	Ebn at Noon		957	994	994	980	955	947	917	916	936	951	927	922
	Edh at Noon		66	78	93	105	109	108	112	107	92	74	67	62
All-Sky Solar Radiation	RadAvg		2.39	3.33	4.59	5.46	6.13	7.20	6.92	6.08	5.13	3.63	2.59	2.03
	RadStd		0.11	0.20	0.33	0.37	0.51	0.37	0.28	0.18	0.28	0.26	0.15	0.13

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page